

Eclipse Dataspace Protocol Integration

Stefan Wiedemann - Senior Software Engineer

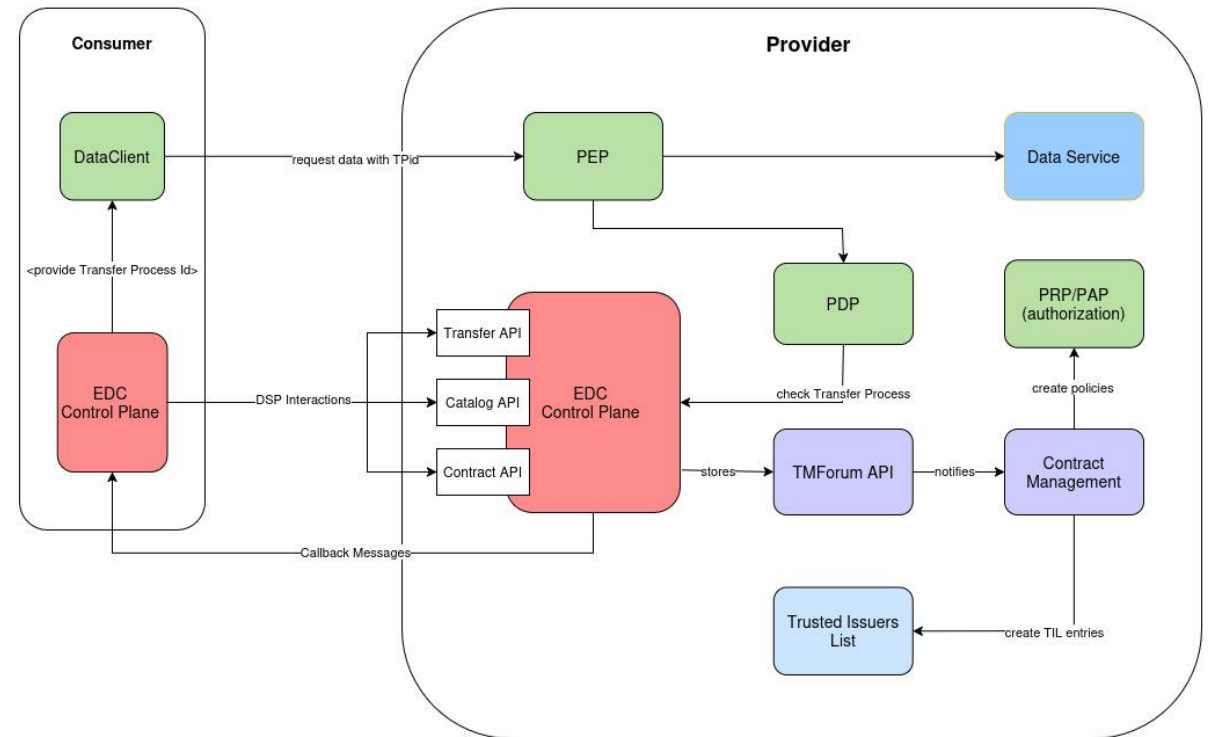


Basic principles

- Extension of EDC Implementation through its SPI's
 - no custom DSP implementation
 - future compatibility easier to ensure
- Separate Data Space Protocol and Decentralized Claims Protocol
 - allows the Auth-Framework to support DCP and OpenId4VP
- Storage of DSP Entities(Catalogs, Offerings, Agreements etc.) as corresponding entities in TMForum
 - automatic integration to the FIWARE DSC Contract Management

Integration of the DSP

- EDC Control Plane provides Transfer, Catalog and Contract API
- Catalog API is filled with Offers from the TMForum API
- Contract Negotiation happens between the Control Planes, all changes are reflected in TMForum “Quote” and “Agreement” objects
- Transfer Process is controlled through the Transfer API, the resulting Process ID is provided to the Data Client*
- Data Client accesses Data through PEP, that validate the Transfer Process



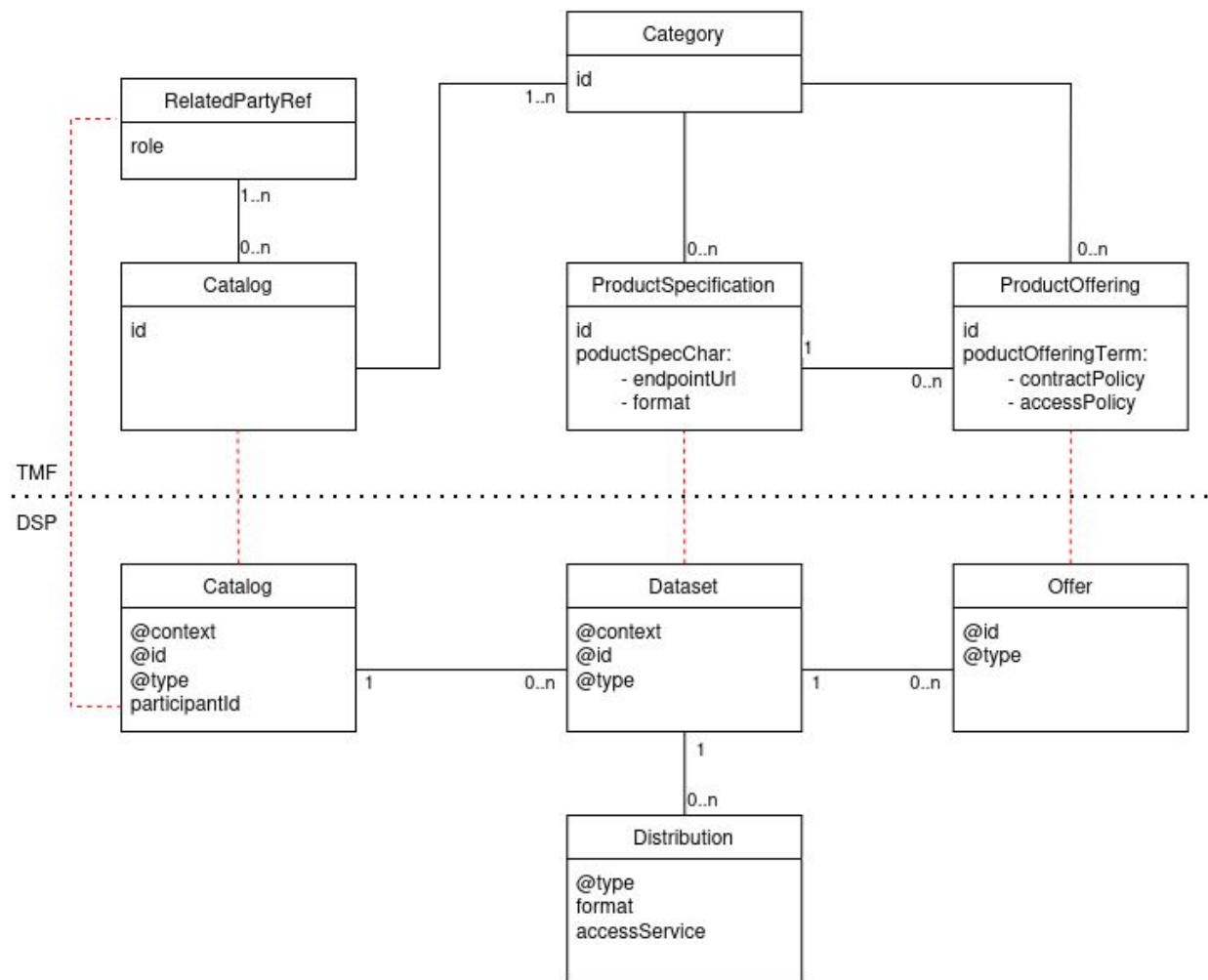
Mapping - Catalog

- DSP Catalog:
 - <https://docs.internationaldataspaces.org/ids-knowledgebase/dataspace-protocol/catalog/catalog.protocol>
- TMForum Catalog
 - <https://www.tmforum.org/oda/open-apis/directory/product-catalog-management-api-TMF620/v4.1>
- Both APIs specify how Data(Datasets, Services) Offerings are defined

Mapping - Catalog

- TMForum Catalog is directly mapped to DSP Catalogs
 - TMForum allows multiple Catalogs per Participant, thus the “DSP Catalog” has to be selected
- DSP “participantId” maps to the more powerful “RelatedPartyRef” with role “Provider”
- DSP Dataset is mapped to TMForum ProductSpecification
 - ProductSpecification is more generic, thus needs to contain some DSP mandatory fields as “productSpecCharacteristics”
- DSP Offer is mapped to TMForum ProductOffering
 - TMForum ProductOfferings reference ProductSpecifications, while DSP Datasets contain the Offer
 - ProductOfferings contain the Policies to be mapped into the Offers
- In TMForum, ProductSpecification and ProductOffering are linked to Catalogs through Categories
 - a “DSP” category could be used for Catalog-Inclusion

Mapping - Catalog



Mapping - Contract Negotiation

- DSP Contract Negotiation:
 - <https://docs.internationaldataspaces.org/ids-knowledgebase/dataspace-protocol/contract-negotiation/contract-negotiation.protocol>
- TMForum Quote API
 - <https://www.tmforum.org/oda/open-apis/directory/quote-management-api-TMF648/v4.0>
- Both APIs provide mechanisms for negotiating the concrete terms of an agreement, based on existing offerings

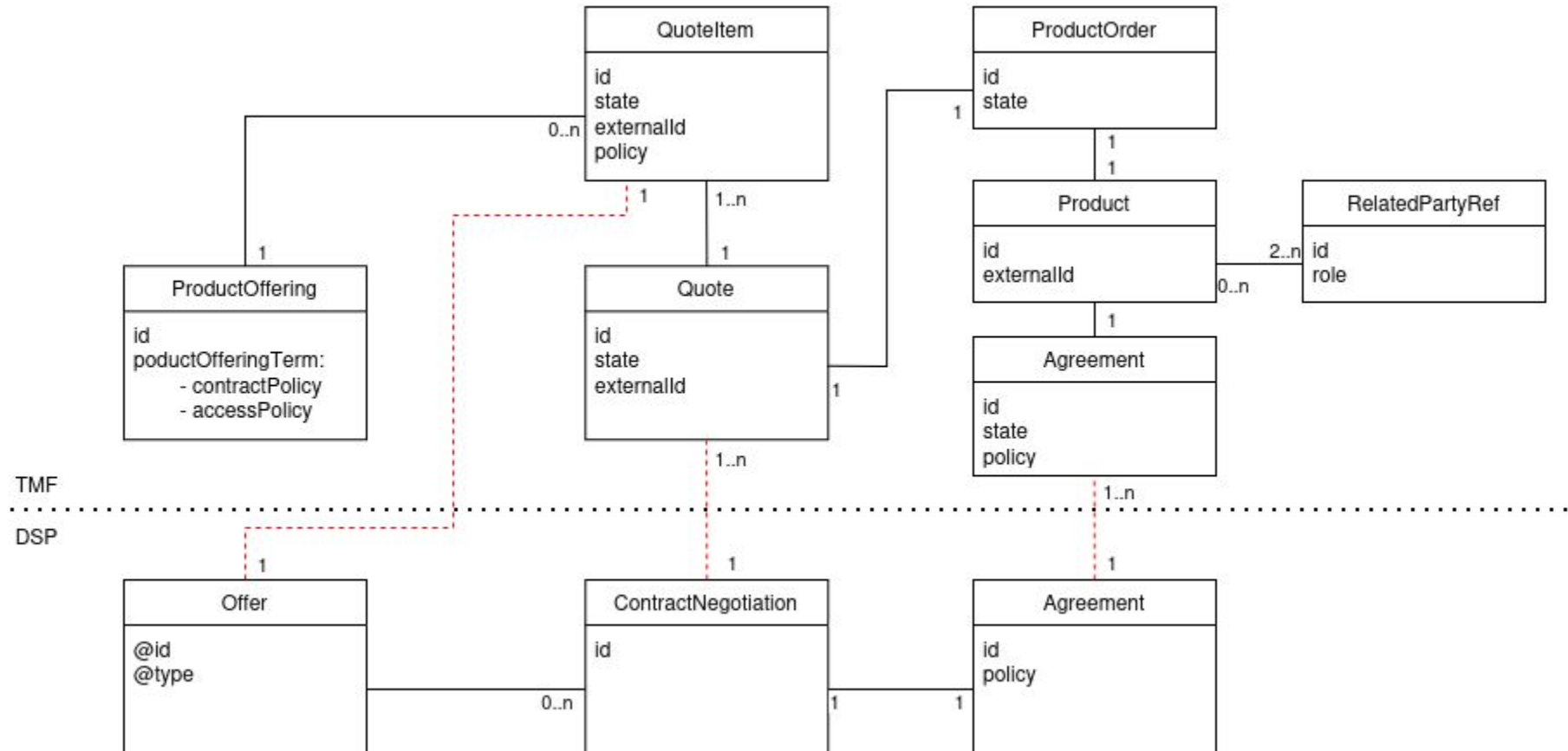
Mapping - Contract Negotiation

- Contract Negotiation is represented by TMForum Quote
 - negotiations in TMForum (can) consist of multiple Quotes, thus a negotiation is mapped by multiple Quotes sharing the same “externalId”(id of the negotiation)
 - implementations have to ensure, that per negotiation only one Quote in an active state(e.g. not “cancelled” or “rejected”) exists
- DSP Negotiation Offer is mapped to TMForum QuoteItem
 - initial Offer is taken from the ProductOffering
 - QuoteItem references the ProductOffering it was originally based on
 - contains the currently offered/counter-offered policy

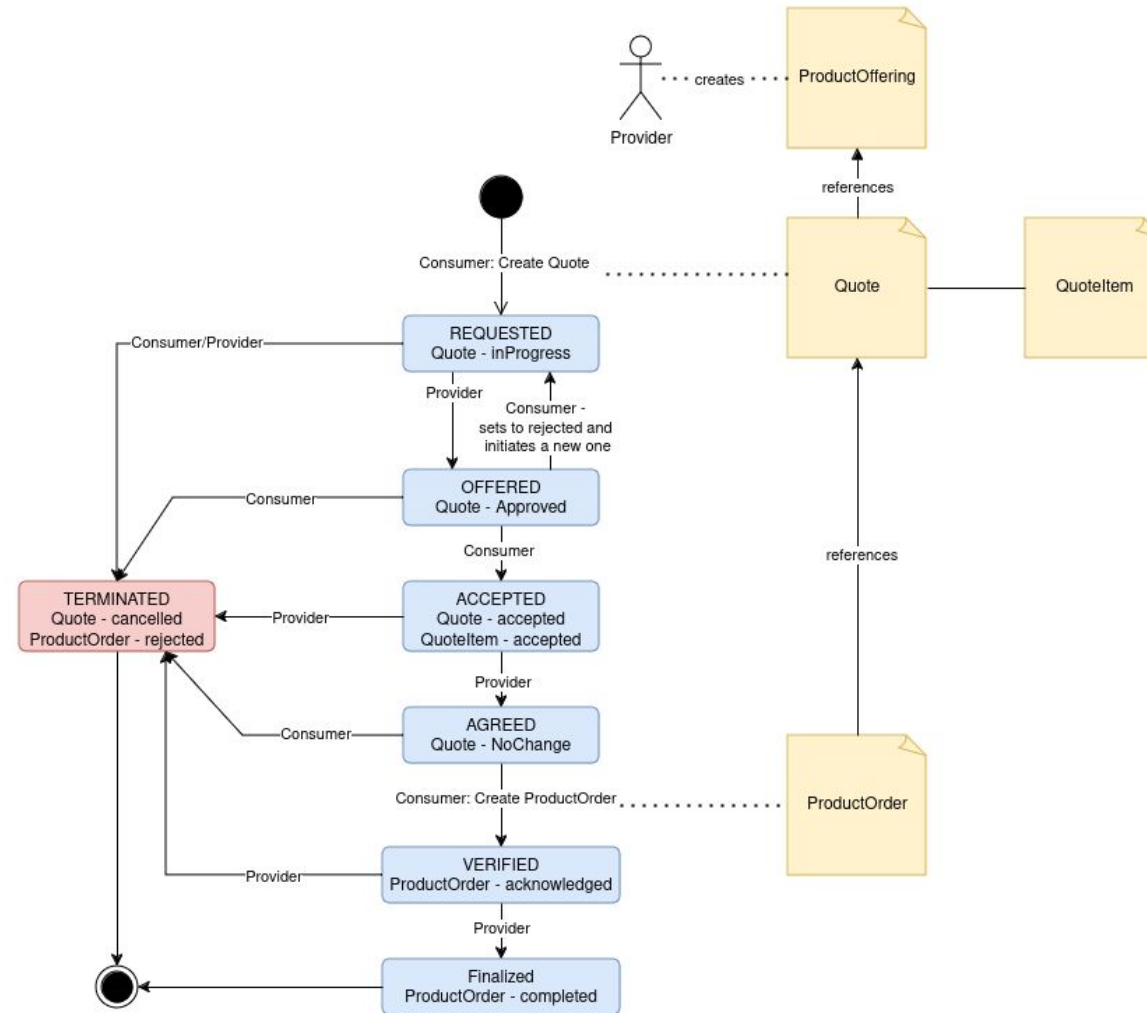
Mapping - Contract Negotiation

- DSP Agreement maps to TMForum Agreement
 - according to the TMForum, a negotiation results in a ProductOrder(state “completed”), a Product(representing the materialization of the ProductSpecification) and the Agreement(referencing the Agreement)
 - ProductOrder references the agreed on Quote(state “accepted”)
 - Product references the ProductOrder, contains RelatedPartyRefs to “Provider” and “Consumer”
 - Agreement references the Product, contains the agreed on Policy

Mapping - Contract Negotiation

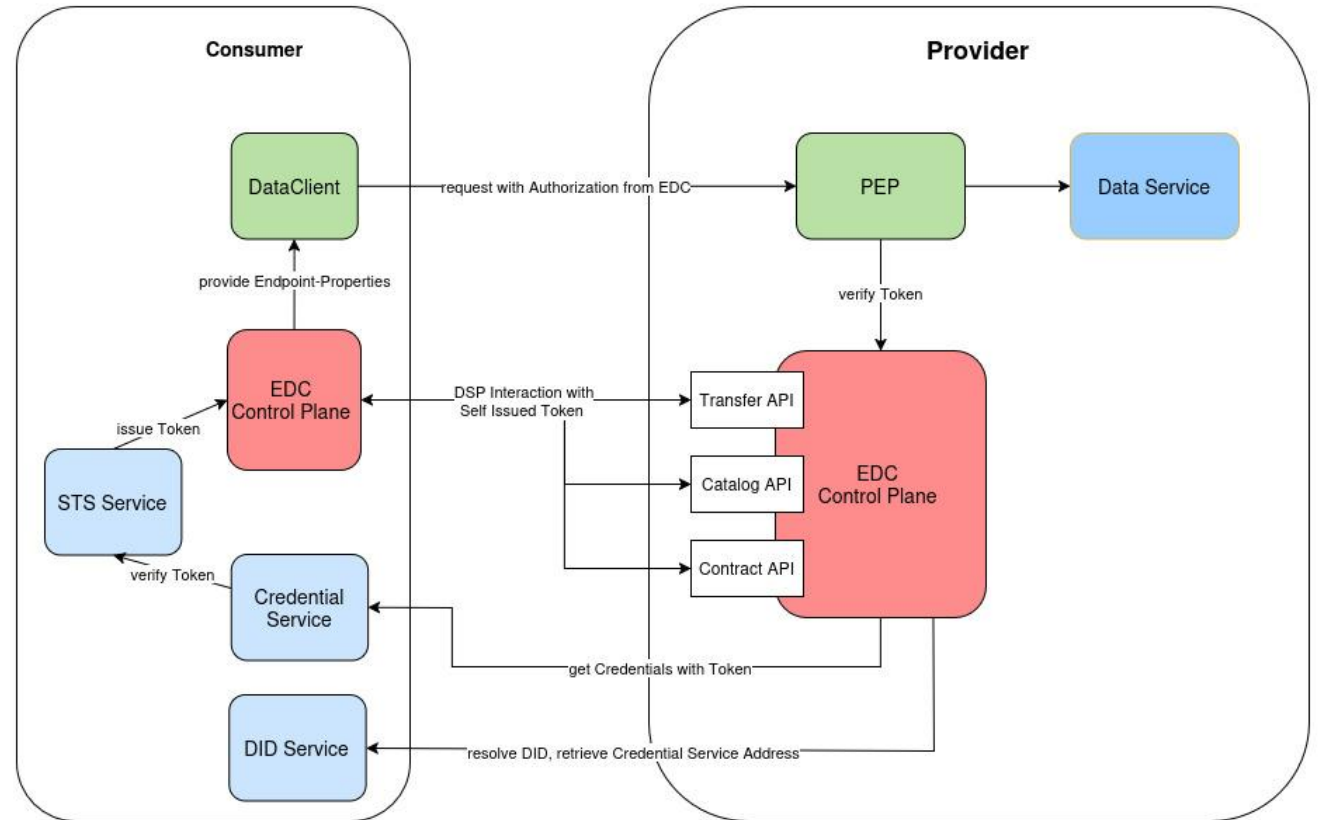


Mapping - Contract Negotiation



DCP Integration

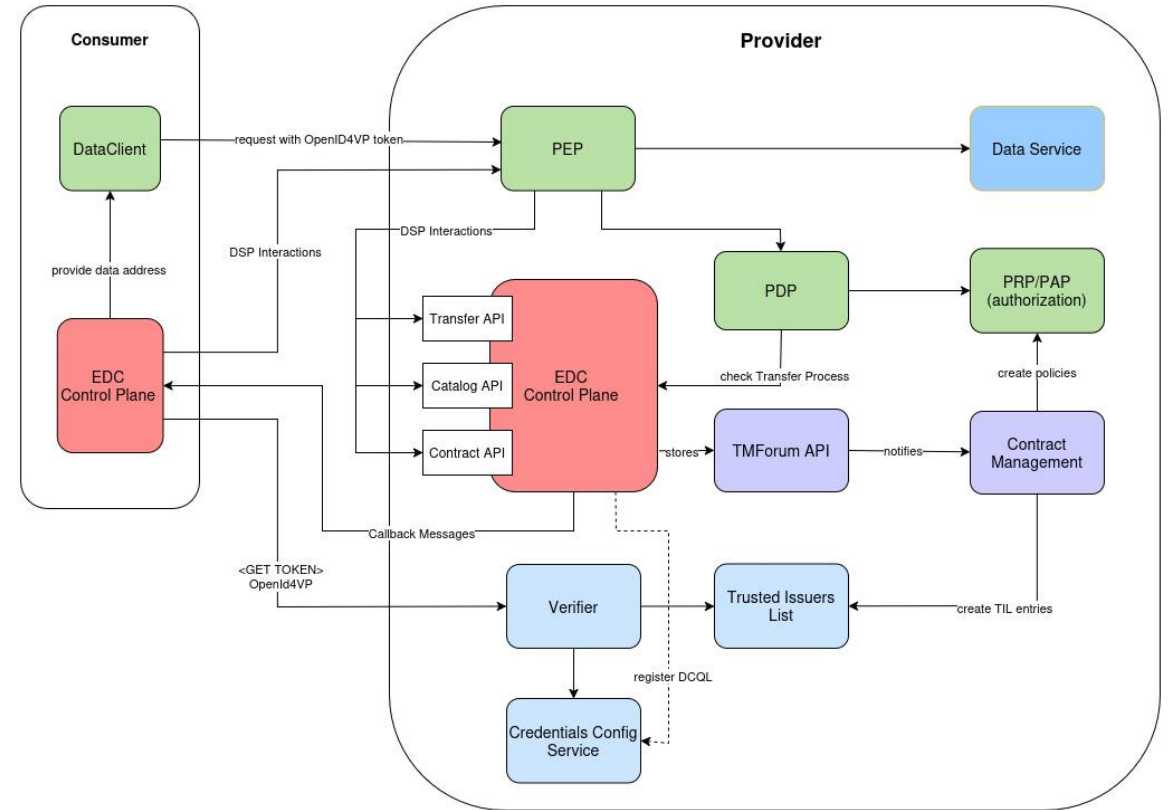
- Integration of the Eclipse Credential Service and STS Service into the Consumer
- Extension of the DID to provide Credential Service address
- Usage of EDC standard Implementation for DCP, following the specified flows
- as a result of the Transfer Process, EDC provides the Endpoint-Properties to the Data Client
- Data Client requests at PEP, which verifies the Token at the EDC Control Plane



DSP with OpenID4VP

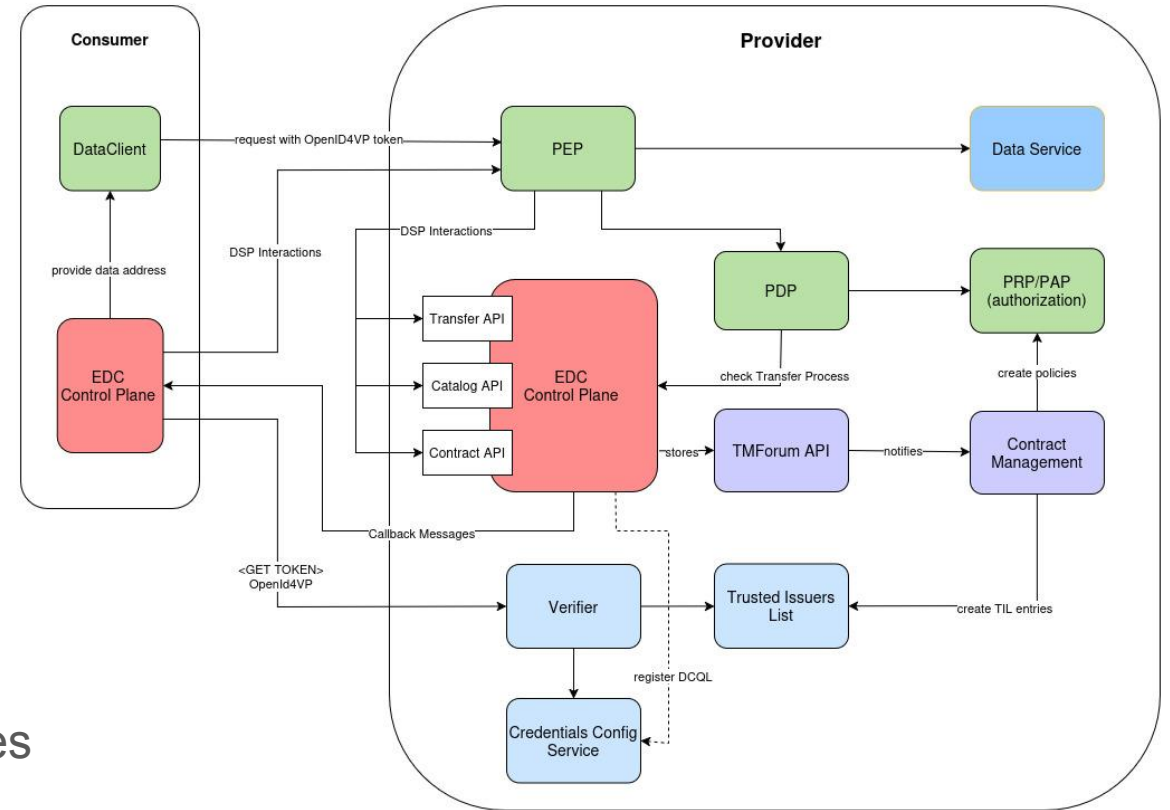
- Access to the Control Plane happens through the PEP, including an OID4VP generated JWT
- Credentials are requested through DCQL by the Verifier, Control Plane generates the required DCQL
- DSP interactions follow the standard implementation
- when Transfer Process is started, Control Plane provides a Data Address, containing the Transfer Process ID:

<https://my-provider.org/<T-PID>/ngsi-ld/v1/entities>



DSP with OpenID4VP

- Data Client requests the Data Address
 - receives a 401 response
 - follows standard OIDC-flow, requests a JWT through OpenID4VP
 - Data Client sends request containing the JWT, PEP validates JWT
 - PDP check Transfer Process ID and Organization ID from JWT at Control Plane
 - forwards request to Data Service
- + interaction can be carried out by different Identities
- + allows direct integration of Humans in the loop
- + uses standards supported by the EU regulation and its Wallet implementations



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Juanjo Hierro

FIWARE TSC chairman

juanjose.hierro@gmail.com

